

Assignment 1.1

GitHub to EC2 Deployment using Jenkins Pipeline

Objective

Deploy a static website from a GitHub repository to an AWS EC2 Linux VM using a Jenkins Pipeline (Declarative). The pipeline checks out the GitHub repository, installs and starts Apache HTTP Server if required, copies the website files to /var/www/html, and verifies that the website is accessible through the EC2 public IP.

Difference from Assignment 1: Assignment 1 used a Jenkins Freestyle project with an Execute shell build step. Assignment 1.1 uses a Jenkins Pipeline with a Jenkinsfile, where each task is a separate stage.

Architecture / Flow

GitHub Repository → Jenkins Pipeline Job → Checkout Stage → Jenkins Workspace → Install Apache Stage → Start Apache Stage → Deploy Website Stage → /var/www/html → Apache :80 → Browser

Item	Value
Source	GitHub repository
Repository	https://github.com/Shantanumajan6/vel-app.git
Branch	master
Jenkins Job	GitHub-Apache-Pipeline
Jenkins Type	Pipeline (Declarative)
Destination	AWS EC2 Linux VM
Web Server	Apache HTTP Server (httpd)
Document Root	/var/www/html
HTTP Port	80

Jenkins Pipeline Deployment Steps

1. Verify the Linux VM

Connect to the EC2 VM using SSH.

```
ssh -i "key_pair.pem" ec2-user@ec2-52-66-250-166.ap-south-1.compute.amazonaws.com
```

Note: Apache installation and service management will be performed by the Jenkins Pipeline.

```
• MobaXterm Personal Edition v26.3 •
(X server, SSH client and network tools)

► Your computer drives are accessible through the /drives path
► Your DISPLAY is set to 192.168.1.38:0.0
► When using SSH, your remote DISPLAY is automatically forwarded
► Each command status is specified by a special symbol (✓ or ✕)

• Important:
This is MobaXterm Personal Edition. The Professional edition
allows you to customize MobaXterm for your company: you can add
your own logo, your parameters, your welcome message and generate
either an MSI installation package or a portable executable.
We can also modify MobaXterm or develop the plugins you need.
For more information: https://mobaxterm.mobatek.net/download.html

2026-10-03 11:43.06 /home/mobaxterm cd Desktop/aws_key/

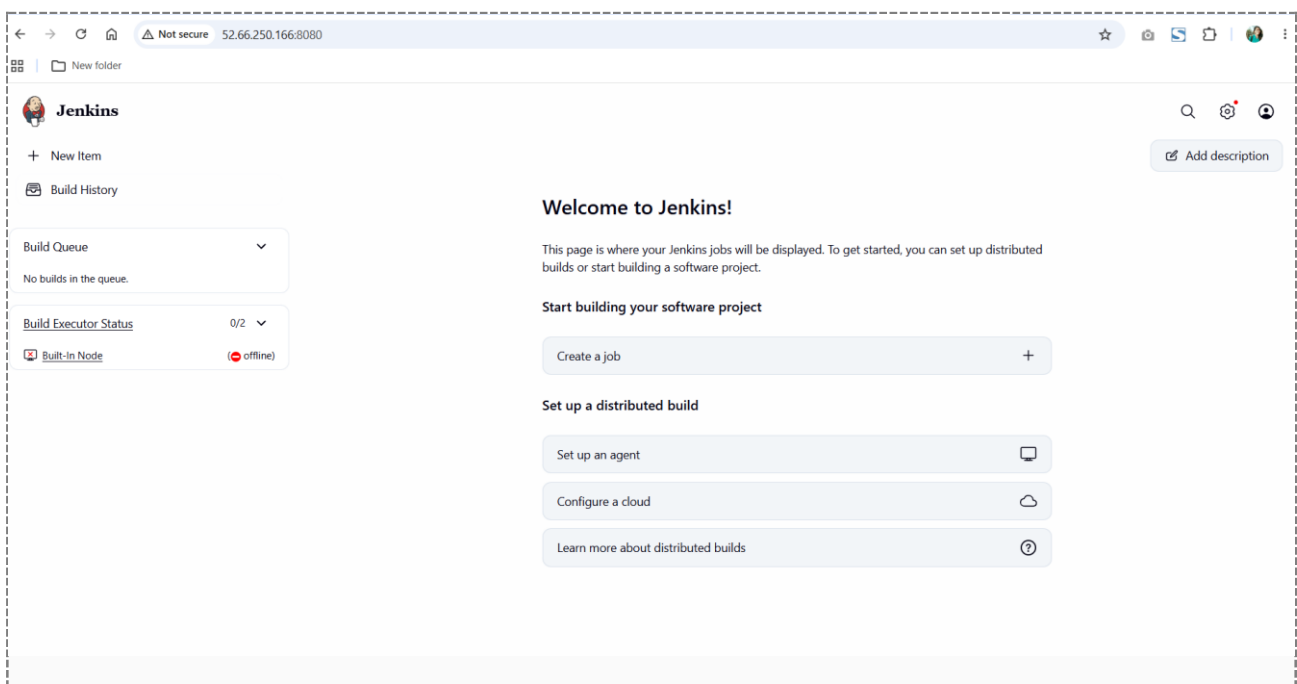
2026-10-03 12:29.56 /home/mobaxterm/Desktop/aws_key ssh -i "key_pair.pem" ec2-user@ec2-52-66-250-166.ap-south-1.compute.amazonaws.com
X11 forwarding request failed on channel 0

#####
##### Amazon Linux 2023
#####
##### https://aws.amazon.com/linux/amazon-linux-2023
#####
#####
#####
#####
#####
#####

Last login: Sat Oct 3 02:54:29 2026 from 103.115.201.70
[ec2-user@ip-172-31-1-51 ~]$ sudo su -
Last login: Sat Oct 3 02:55:27 UTC 2026 on pts/1
[root@ip-172-31-1-51 ~]#
```

2. Install/Verify Jenkins

If Jenkins is already installed on your VM, verify it is running and open the Jenkins dashboard in the browser (<http://52.66.250.166:8080/>).



3. Install and Verify Git

```
yum install git -y
git --version
```

Git must be available on the Jenkins VM so that Jenkins can check out the GitHub repository.

```
[root@ip-172-31-1-51 ~]# git --version
git version 2.50.1
[root@ip-172-31-1-51 ~]#
```

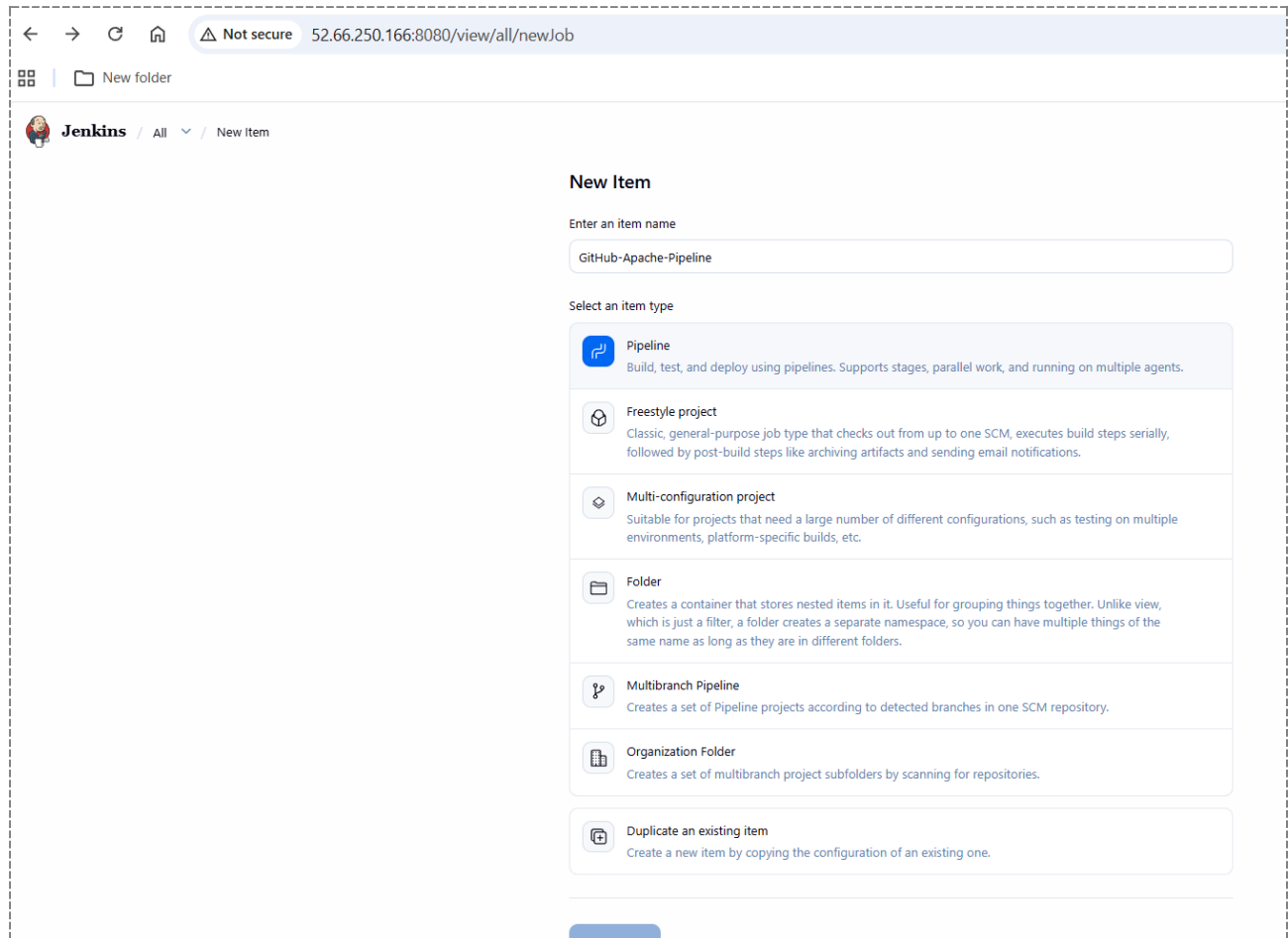
4. Verify Pipeline Plugin

Go to: Manage Jenkins → Plugins → Installed plugins. Confirm the Pipeline plugin (workflow-aggregator) and Git plugin are installed. If not, install them and restart Jenkins.

5. Create Jenkins Pipeline Project

In Jenkins:

1. Click New Item
2. Enter: GitHub-Apache-Pipeline
3. Select Pipeline
4. Click OK



6. Configure the Pipeline

Scroll to the Pipeline section of the job configuration and set:

1. Definition: Pipeline script
2. Script: paste the Jenkinsfile given in Step 7
3. Click Save

Note: Manual git clone is not required. The Checkout stage of the pipeline performs the Git checkout from <https://github.com/Shantanumajan6/vel-app.git> (branch master). The repository contains index.html and dev.html.

7. Jenkinsfile

Use the following Declarative Pipeline script:

```

pipeline {
    agent any

    stages {
        stage('Checkout') {
            steps {
                git branch: 'master',
                    url: 'https://github.com/Shantanumajan6/vel-app.git'
            }
        }

        stage('Install Apache') {
            steps {
                sh '''
                echo "==== Installing Apache ====="
                sudo yum install httpd -y
                '''
            }
        }

        stage('Start Apache') {
            steps {
                sh '''
                echo "==== Starting Apache ====="
                sudo systemctl enable httpd
                sudo systemctl start httpd
                '''
            }
        }

        stage('Deploy Website') {
            steps {
                sh '''
                echo "==== Deploying Website ====="
                sudo cp -r ./* /var/www/html/
                sudo chown -R apache:apache /var/www/html
                sudo chmod -R 755 /var/www/html
                ls -la /var/www/html/
                '''
            }
        }

        stage('Verify Deployment') {
            steps {
                sh '''
                echo "==== Apache Status ====="
                sudo systemctl status httpd --no-pager
                echo "==== Testing Website ====="
                curl -I http://localhost
                '''
            }
        }
    }
}

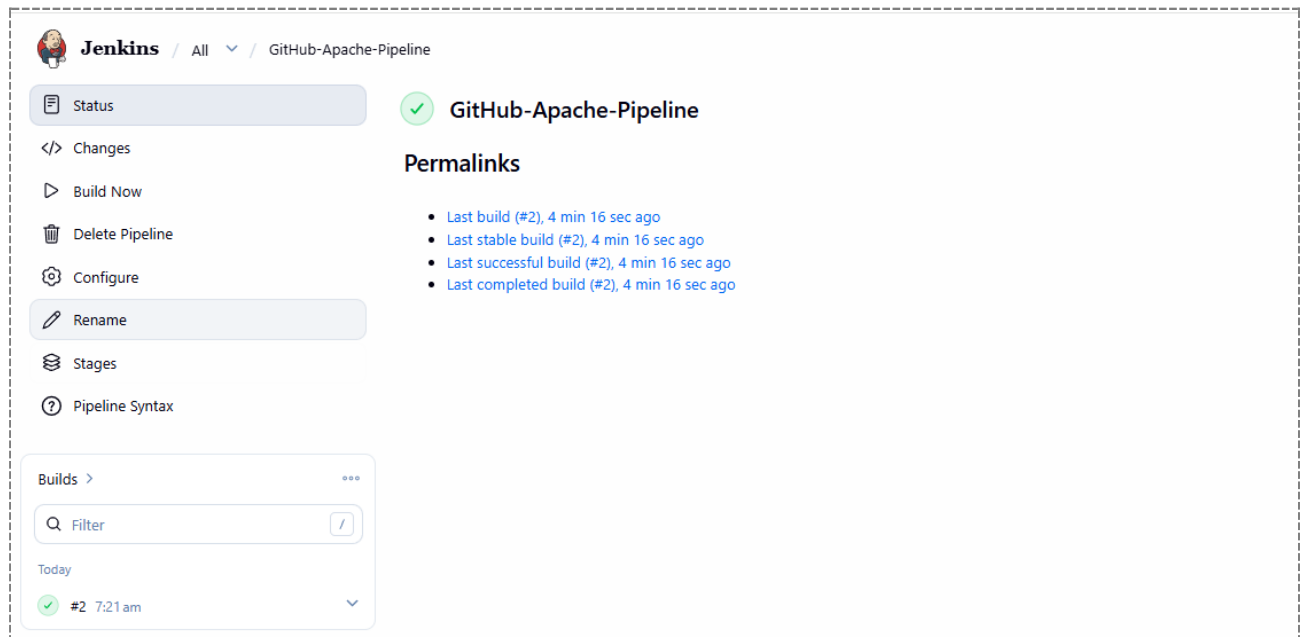
```

Stages: Checkout → Install Apache → Start Apache → Deploy Website → Verify Deployment.

Note: The Jenkins user must be allowed to run sudo without a password on the EC2 VM (for example, via an entry in /etc/sudoers.d/). Otherwise the sudo commands in the stages will fail.

8. Run the Pipeline (Build Now)

Click Build Now on the job page. Open the build and check the Stage View to confirm every stage is green.



The screenshot displays the Jenkins interface for the 'GitHub-Apache-Pipeline' job. The top navigation bar shows 'Jenkins / All / GitHub-Apache-Pipeline'. On the left sidebar, the 'Build Now' button is highlighted. The main content area shows a green checkmark icon next to the job name 'GitHub-Apache-Pipeline'. Below this, the 'Permalinks' section lists four links: 'Last build (#2), 4 min 16 sec ago', 'Last stable build (#2), 4 min 16 sec ago', 'Last successful build (#2), 4 min 16 sec ago', and 'Last completed build (#2), 4 min 16 sec ago'. At the bottom, the 'Builds' section shows a list of builds with a filter input and a dropdown menu. The first build is highlighted, showing a green checkmark, the number '#2', and the time '7:21 am'.

Open Console Output and confirm the build ends with Finished: SUCCESS.

Status

<> Changes

Console Output

Delete build #2

Git Build Data

Pipeline Overview

Edit build information

Restart from Stage

Replay

Pipeline Steps

Workspaces

Console

Download

Copy

View as plain text

```
Started by user Admin
[Pipeline] Start of Pipeline
[Pipeline] node
Still waiting to schedule task
Waiting for next available executor
Pausing (shutting down)
Resuming build at Sat Oct 03 07:23:12 UTC 2026 after Jenkins restart
Running on Jenkins in /var/lib/jenkins/workspace/GitHub-Apache-Pipeline
Ready to run at Sat Oct 03 07:23:12 UTC 2026
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/GitHub-Apache-Pipeline/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/Shantanujand/vel-app.git # timeout=10
Fetching upstream changes from https://github.com/Shantanujand/vel-app.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/Shantanujand/vel-app.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/master^{commit} # timeout=10
Checking out Revision 4384811c9c4e424b56ac08f8db67686302fa2 (refs/remotes/origin/master)
> git config core.sparsecheckout # timeout=10
> git checkout -f 4384811c9c4e424b56ac08f8db67686302fa2 # timeout=10
> git branch -a -v --no-abbrev # timeout=10
> git branch -D master # timeout=10
> git checkout -b master 4384811c9c4e424b56ac08f8db67686302fa2 # timeout=10
Commit message: "Added post build actions"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Install Apache)
[Pipeline] sh
+ echo '==== Installing Apache ====='
==== Installing Apache =====
+ sudo yum install httpd -y
Last metadata expiration check: 4:17:22 ago on Sat Oct 3 03:05:55 2026.
Package httpd-2.4.68-1.amzn2023.0.1.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Start Apache)
[Pipeline] sh
+ echo '==== Starting Apache ====='
==== Starting Apache =====
+ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
+ sudo systemctl start httpd
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy Website)
[Pipeline] sh
+ echo '==== Deploying Website ====='
==== Deploying Website =====
+ sudo cp -r ./Jenkinsfile ./dev.html ./index.html /var/www/html/
+ sudo chown -R apache:apache /var/www/html
+ sudo chmod -R 755 /var/www/html
+ ls -la /var/www/html/
total 12
drwxr-xr-x. 5 apache apache 89 Oct 3 07:23 .
drwxr-xr-x. 4 root root 33 Oct 3 03:12 ..
-rwxr-xr-x. 1 apache apache 902 Oct 3 07:23 Jenkinsfile
drwxr-xr-x. 2 apache apache 24 Oct 3 06:19 R1
drwxr-xr-x. 2 apache apache 24 Oct 3 06:19 R2
drwxr-xr-x. 2 apache apache 24 Oct 3 06:19 R3
-rwxr-xr-x. 1 apache apache 27 Oct 3 07:23 dev.html
-rwxr-xr-x. 1 apache apache 30 Oct 3 07:23 index.html
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Verify Deployment)
[Pipeline] sh
+ echo '==== Apache Status ====='
==== Apache Status =====
+ sudo systemctl status httpd --no-pager
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-10-03 03:14:37 UTC; 4h 0min ago
     Docs: man:httd.service(8)
  Main PID: 28835 (httpd)
    Status: "Total requests: 36; Idle/Busy workers 100/0;Requests/sec: 0.00241; Bytes served/sec: 2.8/sec"
       Tasks: 230 (limit: 1059)
      Memory: 18.4M
         CPU: 16.844s
      CGroup: /system.slice/httpd.service
              └─28835 /usr/sbin/httpd -DFOREGROUND
                └─28858 /usr/sbin/httpd -DFOREGROUND
                  └─28860 /usr/sbin/httpd -DFOREGROUND
                    └─28861 /usr/sbin/httpd -DFOREGROUND
                      └─28869 /usr/sbin/httpd -DFOREGROUND
                        └─29135 /usr/sbin/httpd -DFOREGROUND

Oct 03 03:14:37 ip-172-31-1-51.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Oct 03 03:14:37 ip-172-31-1-51.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Oct 03 03:14:37 ip-172-31-1-51.ap-south-1.compute.internal httpd[28835]: Server configured, listening on: port 80
+ echo '==== Testing Website ====='
==== Testing Website =====
+ curl -I http://localhost
% Total % Received % Xferd Average Speed Time Time Time Current
                               Dload Upload Total Spent Left Speed

  0   0  0  0  0  0  0  0  0  0  0  0  0
  0  30  0  0  0  0  0  0  0  0  0  0
  0  30  0  0  0  0  0  0  0  0  0  0
  0  30  0  0  0  0  0  0  0  0  0  0

HTTP/1.1 200 OK
Date: Sat, 03 Oct 2026 07:23:20 GMT
Server: Apache/2.4.68 (Amazon Linux)
Last-Modified: Sat, 03 Oct 2026 07:23:20 GMT
ETag: W/"1e-65ceae85f87af7"
Accept-Ranges: bytes
Content-Length: 30
Content-Type: text/html; charset=UTF-8

[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

One Jenkins pipeline run now demonstrates the full flow: Git checkout → Apache installation → Apache startup → website deployment → permissions → verification.

9. Verify Deployment on EC2 VM

After a successful pipeline run, SSH into the VM and run:

```
ls -la /var/www/html/
```

Expected files: index.html, dev.html

Then:

```
curl -I http://localhost
```

Expected: HTTP/1.1 200 OK

```
[root@ip-172-31-1-51 html]# ls -l
total 12
-rwxr-xr-x. 1 apache apache 902 Oct  3 07:23 Jenkinsfile
drwxr-xr-x. 2 apache apache 24 Oct  3 06:19 R1
drwxr-xr-x. 2 apache apache 24 Oct  3 06:19 R2
drwxr-xr-x. 2 apache apache 24 Oct  3 06:19 R3
-rwxr-xr-x. 1 apache apache 27 Oct  3 07:23 dev.html
-rwxr-xr-x. 1 apache apache 30 Oct  3 07:23 index.html
[root@ip-172-31-1-51 html]#
```

10. Verify Website from Browser

Open: <http://52.66.250.166/index.html>

You can also test: <http://52.66.250.166/index.html>

